

ARRAYS

Definition

"Arrays store multiple values in a single variable."

Purpose

Arrays manage collections of data efficiently

Syntax

```
let fruits = ["Apple", "Mango"];
```

Access Array Elements

```
console.log(fruits[0]);
```

Array Methods

1. push()

Adds elements at end.

```
fruits.push("Banana");
```

2. pop()

Removes last element.

fruits.pop();

3. shift()

Removes first element.

fruits.shift();

4. unshift()

Adds element at beginning.

fruits.unshift("Orange");

5. map()

Transforms array.

numbers.map(num => num * 2);

6. filter()

Filters data.

numbers.filter(num => num > 5);

7. reduce()

Reduces array to single value.

numbers.reduce((a,b) => a+b);

8. Sorting Arrays

numbers.sort((a,b) =>

a-b);

* Real World Use

Arrays are

used in:

- Product lists
- Cart systems
- notifications
- social media feeds
- dashboards

Important Points

- Arrays start from index 0.
- map/filter/reduce are modern methods.
- Arrays are dynamic in JavaScript

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